

Supplementary material for: A paid application standardises what large language models read about insulin pump settings and leaves what they decide to chance: reproducibility of an automated insulin delivery settings review across eleven models

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Table S1. Model identifiers as requested and as returned by each provider

The requested identifier is the string sent in the API call; the served identifier is the model field the provider returned in each response, with the number of turns reporting it. Dates are the first and last conversation start dates (UTC).

Model (paper)	Provider and route	Requested identifier	Served identifier (turns)	Conversations	Dates
Gemini 3.6 Flash	Google, Generative Language API	gemini-3.6-flash	gemini-3.6-flash (204)	50	2026-08-21 to 2026-08-21
Gemini 3.1 Pro Preview	Google, Generative Language API	gemini-3.1-pro-preview	gemini-3.1-pro-preview (203)	50	2026-08-21 to 2026-08-21
Gemini 3.5 Flash-Lite	Google, Generative Language API	gemini-3.5-flash-lite	gemini-3.5-flash-lite (218)	50	2026-08-22 to 2026-08-22
GPT-5.6-sol	OpenRouter, OpenAI-compatible API (reasoning effort medium)	gpt-5.6-sol	gpt-5.6-sol (200)	50	2026-08-21 to 2026-08-21
GPT-5.4-mini	OpenRouter, OpenAI-compatible API (reasoning effort medium)	gpt-5.4-mini	gpt-5.4-mini-2026-03-17 (227)	50	2026-08-22 to 2026-08-22
Claude Opus 5	Anthropic, Messages API	claude-opus-5	claude-opus-5 (95)	20	2026-08-21 to 2026-08-22
Claude Sonnet 5	Anthropic, Messages API	claude-sonnet-5	claude-sonnet-5 (84)	20	2026-08-21 to 2026-08-22
Claude Haiku 4.5	Anthropic, Messages API	claude-haiku-4-5	claude-haiku-4-5-20251001 (244)	50	2026-08-22 to 2026-08-22
DeepSeek V4 Pro	OpenRouter, OpenAI-compatible API	deepseek/deepseek-v4-pro	deepseek/deepseek-v4-pro (217)	50	2026-08-22 to 2026-08-22
Grok 4.6	OpenRouter, OpenAI-compatible API	x-ai/grok-4.6	x-ai/grok-4.6 (201)	50	2026-08-22 to 2026-08-22
Llama 4 Maverick	OpenRouter, OpenAI-compatible API	meta-llama/llama-4-maverick	meta-llama/llama-4-maverick (231)	50	2026-08-22 to 2026-08-22

Table S2. Verification of the reconstructed validator against the application

Twenty-six real model outputs were pasted into the application's result screen on 22 August 2026 and its response recorded; the same texts were then judged by the validator. Sample 04 was pasted while the application held the demonstration package and is reported only for whether fenced JSON parsed. The verdict classes are: accepted; repair prompt (with the clauses the application composed); schema error with no prompt; invalid JSON with no prompt.

Sample	Source conversation	Defect in the paste	Application's response	Validator's verdict	Agree
01	real_gemini35flashlite_50/run_037	fully compliant, should be accepted	accepted	accepted	yes
02	real_gpt54mini_med_50/run_018	current_value not copied verbatim: parameter_decisions.aaps.core.carb_absorption.current_value	repair prompt: copy current_value for aaps.core.carb_absorption	repair prompt: copy current_value for aaps.core.carb_absorption	yes
03	real_gpt54mini_med_50/run_008	unrequested key added inside meal_strategy_summary	accepted	accepted	yes
04	real_llama4maverick_50/run_035	fully compliant JSON wrapped in ``json fences	accepted (fenced JSON parsed; tested against the demonstration package)	parsed (fenced JSON accepted by extractor)	yes
05	real_gemini35flashlite_50/run_026	confidence outside low/medium/high in: analysis_steps.meal_ecarbs_absorption.confidence	accepted	accepted	yes
06	real_deepseek_v4pro_50/run_041	profile_recommendation.focus outside basal/target/cr/isf/dia	repair prompt: profile_recommendation.focus	repair prompt: profile_recommendation.focus	yes
07	real_opus5/run_017	recommendation priority outside low/medium/high/safety	accepted	accepted	yes
08	real_gemini31pro_50/run_024	top-level section(s) missing: issues	accepted	accepted	yes
09	real_gpt54mini_med_50/run_043	sections flagged: analysis_steps.automation_sensitivity_limits.confidence, meal_strategy_summary	accepted	accepted	yes
10	real_deepseek_v4pro_50/run_023	sections flagged: analysis_steps.automation_smb_uam.confidence, profile_recommendation.focus	repair prompt: profile_recommendation.focus	repair prompt: profile_recommendation.focus	yes
11	real_gemini35flashlite_50/run_050	sections flagged: analysis_steps.basal_targets.confidence, analysis_steps.synthesis_plan.confidence, profile_recommendation	schema error, no prompt (application session then crashed)	schema error, no prompt	yes
12	real_deepseek_v4pro_50/run_033	sections flagged: analysis_steps.profile_cr_isf_dia.confidence, parameter_decisions.profile.basal.00_00.current_value, parameter_decisions.profile.basal.05_00.current_value, parameter_decisions.profile.basal.06_00.current_value, parameter_decisions.profile.basal.10_00.current_value, parameter_decisions.profile.basal.16_00.current_value, parameter_decisions.profile.basal.18_00.current_value, parameter_decisions.profile.basal.19_00.current_value, parameter_decisions.profile.cr.00_00.current_value, parameter_decisions.profile.cr.04_00.current_value, parameter_decisions.profile.cr.08_00.current_value, parameter_decisions.profile.cr.16_00.current_value, parameter_decisions.profile.cr.20_00.current_value, parameter_decisions.profile.isf.00_00.current_value, parameter_decisions.profile.target.00_00.current_value, parameter_decisions.profile.target.08_00.current_value, parameter_decisions.profile.target.23_00.current_value, parameter_decisions.profile.dia.current_value	repair prompt: copy current_value for 17 keys	repair prompt: copy current_value for 17 keys	yes
13	real_grok46_50/run_017	sections flagged: analysis_steps.safety_overview, analysis_steps.basal_targets, analysis_steps.profile_cr_isf_dia, analysis_steps.automation_smb_uam, analysis_steps.automation_sensitivity_limits, analysis_steps.meal_bolus_strategy, analysis_steps.meal_ecarbs_absorption, analysis_steps.synthesis_plan	accepted	accepted	yes

Sample	Source conversation	Defect in the paste	Application's response	Validator's verdict	Agree
14	real_gpt54mini_med_50/run_003	sections flagged: meal_strategy_summary.confidence	accepted	accepted	yes
15	real_gemini35flashlite_50/run_015	sections flagged: parameter_decisions.aaps.core.carb_absorption	accepted	accepted	yes
16	real_deepseek_v4pro_50/run_002	sections flagged: parameter_decisions.profile.basal.10_00, parameter_decisions.profile.basal.16_00	schema error, no prompt	schema error, no prompt	yes
17	real_deepseek_v4pro_50/run_035	sections flagged: parameter_decisions.profile.cr.00_00	accepted	accepted	yes
18	real_gemini36flash_50/run_016	sections flagged: parameter_decisions.profile.dia	accepted	accepted	yes
19	real_gpt54mini_med_50/run_042	sections flagged: profile_recommendation	accepted	accepted	yes
20	real_gemini35flashlite_50/run_017	valid JSON followed by a stray closing ``	accepted	accepted	yes
21	real_gemini35flashlite_50/run_028	well-formed except one closing brace missing (root never closes)	invalid JSON, no prompt	invalid JSON, no prompt	yes
22	real_opus5/run_011	several top-level JSON objects in sequence instead of one	invalid JSON, no prompt	invalid JSON, no prompt	yes
23	real_llama4maverick_50/run_012	syntax error inside: Expecting value at char 0	invalid JSON, no prompt	invalid JSON, no prompt	yes
24	real_gemini35flashlite_50/run_025	stray closing `` at the end, no opening fence	invalid JSON, no prompt	invalid JSON, no prompt	yes
25	real_gemini35flashlite_50/run_033	parameter_decisions rows missing/duplicated/extra: parameter_decisions	repair prompt: add profile.cr.16_00, remove profile.cr.06_00	repair prompt: add profile.cr.16_00; remove profile.cr.06_00	yes
26	real_gpt54mini_med_50/run_018 (response to the repair prompt)	current_value still not copied exactly ("5m" for "5 min")	repair prompt: copy current_value for aaps.core.carb_absorption (second time)	repair prompt: copy current_value for aaps.core.carb_absorption	yes

Agreement on verdict class: 25 of 25 informative samples.

Table S3. Whether the rationale for each decision cites a figure from the record

Post hoc analysis over accepted conversations. For each decision row the rationale text was searched for figures with a unit (the same extraction as the citation check) and each figure was checked against the numbers in the package. A change decision is counted as cited when at least one such figure is present in the package. Wilson

95 per cent intervals, with the decision row as the unit, are given for the cited proportion; rows are not independent within a conversation, so the intervals are indicative.

Model	Change decisions	Cited a verified figure, n (per cent; 95 per cent interval)	Cited figures, none verified	No figure in rationale	Keep decisions	Keep decisions citing a verified figure, n (per cent)
Gemini 3.6 Flash	160	154 (96; 92 to 98)	0	6	990	328 (33)
Gemini 3.1 Pro Preview	141	97 (69; 61 to 76)	0	44	1009	88 (9)
Gemini 3.5 Flash-Lite	10	9 (90; 60 to 98)	0	1	887	129 (15)
GPT-5.6-sol	20	17 (85; 64 to 95)	0	3	1130	187 (17)
GPT-5.4-mini	2	0 (0; 0 to 66)	0	2	1102	3 (0)
Claude Opus 5	16	16 (100; 81 to 100)	0	0	237	200 (84)
Claude Sonnet 5	1	1 (100; 21 to 100)	0	0	436	150 (34)
Claude Haiku 4.5	27	27 (100; 88 to 100)	0	0	433	405 (94)
DeepSeek V4 Pro	93	84 (90; 83 to 95)	0	9	1011	105 (10)
Grok 4.6	16	16 (100; 81 to 100)	0	0	1134	615 (54)
Llama 4 Maverick	56	3 (5; 2 to 15)	0	53	956	0 (0)

Table S4. How accepted conversations read the package's bolus-only insulin total

Post hoc analysis. The package's treatment_summary gives total_insulin_u 275.95 over 14 days (19.7 U/day), a sum of the bolus and SMB treatment records only: the package carries no basal delivery data, and its scheduled basal profile integrates to 16.2 U/day. Response text was searched for the derived figure (19.7 or 275.95) in a daily total or TDD context, for statements that the total excludes basal or is bolus only, and for a reconstructed total near 36 U/day. Wilson 95 per cent intervals over accepted conversations.

Model	Accepted	Treated 19.7 U as a daily total or TDD, n (per cent; 95 per cent interval)	Noted the total excludes basal, n (per cent)	Reconstructed about 36 U/day, n (per cent)	Lowered max IOB while calling 19.7 a daily total
Gemini 3.6 Flash	50	49 (98; 90 to 100)	13 (26)	0 (0)	48
Gemini 3.1 Pro Preview	50	16 (32; 21 to 46)	5 (10)	0 (0)	16
Gemini 3.5 Flash-Lite	39	0 (0; 0 to 9)	5 (13)	0 (0)	0
GPT-5.6-sol	50	4 (8; 3 to 19)	23 (46)	2 (4)	0
GPT-5.4-mini	48	0 (0; 0 to 7)	32 (67)	0 (0)	0
Claude Opus 5	11	9 (82; 52 to 95)	10 (91)	11 (100)	7
Claude Sonnet 5	19	2 (11; 3 to 31)	8 (42)	6 (32)	0
Claude Haiku 4.5	20	4 (20; 8 to 42)	19 (95)	8 (40)	0

Model	Accepted	Treated 19.7 U as a daily total or TDD, n (per cent; 95 per cent interval)	Noted the total excludes basal, n (per cent)	Reconstructed about 36 U/day, n (per cent)	Lowered max IOB while calling 19.7 a daily total
DeepSeek V4 Pro	48	5 (10; 4 to 22)	25 (52)	7 (15)	5
Grok 4.6	50	26 (52; 38 to 65)	23 (46)	29 (58)	0
Llama 4 Maverick	44	0 (0; 0 to 8)	0 (0)	0 (0)	0

Table S5. Further package-inherited splits

Post hoc analysis over accepted conversations. Hour of the lows: the package's hourly glucose is keyed both in UTC (time below range peaks at 18.5 per cent at hour 0) and in local time (the same peak at 01:00, BST); responses are classed by where they place the peak, with '00:00 or midnight only' indicating the UTC array read as clock time. Diluted insulin: the record's use of diluted insulin is disclosed only in a free-text profile name inside Profile Switch events; counts give responses mentioning it and responses engaging with its kinetics or absorption. mmol/L: responses quoting the mmol-denominated ISF values from the loop reason strings inside the otherwise mg/dl package.

Model	Accepted	Peak at 01:00 only	Peak at 00:00 or midnight only	Both hours cited	Mentions diluted insulin	Engages with its kinetics	Quotes mmol ISF values
Gemini 3.6 Flash	50	10	1	39	1	0	0
Gemini 3.1 Pro Preview	50	10	4	32	8	0	0
Gemini 3.5 Flash-Lite	39	12	6	5	0	0	0
GPT-5.6-sol	50	10	0	39	4	1	0
GPT-5.4-mini	48	2	2	3	0	0	0
Claude Opus 5	11	1	0	7	3	3	0
Claude Sonnet 5	19	0	5	1	13	4	0
Claude Haiku 4.5	20	2	3	8	12	9	1
DeepSeek V4 Pro	48	14	6	13	3	2	1
Grok 4.6	50	25	3	10	46	42	0
Llama 4 Maverick	44	0	0	0	0	0	0